Bibliography

[1] S. Peter Andersson and Ove Andersson. “Relaxation Studies of Poly(propylene glycol) under High Pressure”. In: *Macromolecules* 31 (1998), p. 2999.

[2] C. A. Angell. “Formation of Glasses from Liquids and Biopolymers”. In: *Science* 267.5206 (1995), p. 1924.

[3] C. A. Angell. “Insights into Phases of Liquid Water from Study of its Unusual Glass-Forming Properties”. In: *Science* 319 (5863 2008), p. 582.

[4] M. F. Ashby and A. L. Greer. “Metallic glasses as structural materials”. In: *Scripta Materialia* 54 (3 2006), p. 321.

[5] A. J. Barlow, J. Lamb, and A. J. Matheson. “Viscous behaviour of supercooled liquids”. In: *Proc. R. Soc. A* 292.1430 (1966), p. 322.

[6] Z. Batschinski. In: *Phys. Chem.* 84 (1913), p. 644.

[7] Bruce J. Berne and Robert Pecora. *Dynamic Light Scattering*. Dover, Mineola NY, 2000.

[8] K. Binder and A. P. Young. “Spin glasses: Experimental facts, theoretical concepts, and open questions”. In: *Rev. Mod. Phys.* 58 (4 1986), p. 801.

[9] D. Bradley. “Glass Inequality”. In: *Materials Today* 21 (2018), p. 4.

[10] P.W. Bridgman. In: *Proc. Am. Acad. Arts. Sci.* 61 (1926), p. 57.

[11] F. Bueche. “Segmental mobility of polymers near their glass temperature”. In: *J. Chem. Phys.* 21.10 (1953), p. 1850.

[12] Manuel Cardona, Ralph V. Chamberlin, and Marx Werner. “Comment on the history of the stretched exponential function”. In: (2007).

[13] Chiara Caronna et al. “Dynamics of nanoparticles in a supercooled liquid”. In: *Phys. Rev. Lett.* 100 (2008), p. 055702.

[14] R. Casalini and C. Roland. “Scaling of the supercooled dynamics and its relation to the pressure dependences of the dynamic crossover and the fragility of glass formers”. In: *Phys. Rev. B* 71.1 (2005), p. 014210.

[15] I. Cibulka and T. Takagi. “P–t data of liquids: summarization and evaluation. 5. aromatic hydrocarbons”. In: *J. Chem. Eng. Data* 44 (1999), p. 411.

[16] Morrel H. Cohen and David Turnbull. “Molecular transport in liquids and glasses”. In: *J. Chem. Phys.* 31.5 (1959), p. 1164.

[17] Richard L. Cook, Chris A. Herbst, and H.E. King. “High-pressure viscosity of glass-forming liquids measured by the centrifugal force diamond anvil cell viscometer”. In: *J. Phys. Chem.* 97 (1993), p. 2355.

[18] Richard Lee Cook. “High pressure viscosity of simple and polymeric fluids”. PhD thesis. Harvard University, 1993.

[19] Richard L. Cook et al. “Pressure and temperature dependent viscosity of two glass forming liquids: Glycerol and dibutyl phthalate”. In: *J. Phys. Chem.* 100.7 (1994), p. 5178.

[20] D. Coslovich and C. M. Roland. “Thermodynamic Scaling of Diffusion in Supercooled Lennard-Jones Liquids”. In: *J. Phys. Chem. B* 112 (2007), p. 1329.

[21] D. W. Davidson and R. H. Cole. “Dielectric relaxation in glycerol, propylene glycol, and n-Propanol”. In: *J. Chem. Phys.* 19.12 (1951), p. 1484.

[22] Pablo G. Debenedetti and Gene Stanley. “Supercooled and Glassy Water”. In: *Physics Today* 56 (6 2003), p. 60.

[23] Pablo G. Debenedetti and Frank H. Stillinger. “Supercooled liquids and the glass transition”. In: *Nature* 410 (2001), p. 259.

[24] P. Debye. *Polar Molecules*. Dover, New York, 1945.

[25] R. Donald and Pardeep P. Fulay. *Essentials of Materials Science & Engineering*. Cengage Learning, 2008.

[26] Austin K. Doolittle. “Studies in Newtonian Flow. II. The dependence of the viscosity of liquids on free-space”. In: *J. App. Phys.* 22.12 (1951), p. 1471.

[27] Henry Eyring. “Viscosity, Plasticity, and Diffusion as Examples of Absolute Reaction Rates”. In: *J. Chem. Phys.* 4 (1936), p. 283.

[28] Hans Forsman, Per Anderson, and Gunnar Backström. “Dielectric relaxation of glycerol and n-propyl alcohol at high pressure”. In: *J. Chem. Soc.* 82 (1986), p. 857.

[29] J. Frenkel. *Kinetic Theory of Liquids*. Oxford University Press, 1946. Chap. 5.

[30] Raymond M. Fuoss and John G. Kirkwood. “Electrical Properties of Solids. VIII. Dipole Moments in Polyvinyl Chloride-Diphenyl Systems”. In: *J. Am. Chem. Soc.* 63 (1941), p. 385.

[31] R. H. Geils and R. C. Keezer. “Small-volume, inclined, falling-ball viscometer”. In: *Rev. Sci. Instrum.* 48.7 (1977), p. 783.

[32] L.S. Grattan. *Optical Fiber Sensor Technology: Advanced Applications–Bragg Gratins and Distributed Sensors*. Springer Science & Business Media, 2013.

[33] A. Lindsay Greer. “Metallic Glasses”. In: *Science* 267.5206 (1995), p. 1947.

[34] The Elliot Group. *Ab initio study of vibrational and structural properties of polyol glasses*. 2013. URL: https://www.ch.cam.ac.uk/group/elliott/research/organic-glasses (visited on 02/19/2019).

[35] A. H.E.Brodin and E. A. Rossler. “Depolarized light scatting study of glycerol”. In: *Eur. Phys. J. B* 44 (2005), p. 3.

[36] D Heiman, D Hamilton, and R W. Hellwarth. “Brillouin scattering measurements on optical glasses”. In: *Phys. Rev. B* 19 (1979), p. 12.

[37] C. A. Herbst et al. “Dynamic light scattering measurments of highpressure viscosity utilizing a diamond anvil cell”. In: *J. Appl. Phys.* 72 (1992), p. 838.

[38] Ian M. Hodge. “Adam-Gibbs formulation of enthalpy relaxation near the glass transition”. In: *J. Res. Natl. Inst. Stand. Technol.* 102.2 (1997), p. 195.

[39] Leon Van Hove. “Correlations in space and time and Born approximation scattering in systems of interacting particles”. In: *Phys. Rev.* 95.1 (1954), p. 249.

[40] R. M. Hubbard and G. G. Brown. In: *Ind. Eng. Chem.* 15 (1943), p. 212.

[41] John David Jackson. *Classical Electrodynamics*. Third. John Wiley & Sons, Inc., 1999.

[42] Xin Jiang et al. “Investigation on germanium oxide-based glasses for infrared optical fibre development”. In: *Optical Materials* 31 (2009), p. 1701.

[43] Gyan P. Johari. “Introduction to the Glassy State in the Undergraduate Curriculum”. In: *J. Chem. Educ.* 51 (1974), p. 23.

[44] Gyan P. Johari and Martin Goldstein. “Viscous liquids and the glass transition. II. Secondary RElaxations in Glasses of Rigid Molecules”. In: *J. Chem. Phys.* 53 (1970), p. 2372.

[45] K. Jurkiewicz et al. “Modelling of glass-like carbon structure and its experimental verification by neutron and X-ray diffraction”. In: *J. Appl. Cryst.* 50 (2017), p. 36.

[46] L. G. Khvostantsev, V. N. Slesarev, and V. V. Brazhkin. “Toroid type high-pressure device: history and prospects”. In: *High Press. Res.* 24 (2004), p. 371.

[47] L. I. Komarov and I. Z. Fisher. “Theory of Rayleigh scattering of light in liquids”. In: *JETP* 16.5 (1962), p. 1358.

[48] Z. W. Kyaw and N. Menon. “Glass transition of glycerol in the volume-temperature plane”. In: *Phys. Rev. E* 73 (2006), p. 040501.

[49] L.D. Landau, E.M. Lifshitz, and L.P. Pitaevskii. *Electrodynamics of Continuous Media*. Second. Vol. 8. Elsevier Butterworth-Heinemann, Burlington MA, 2004.

[50] W. T. Laughlin. PhD thesis. Massachusetts Institute of Technology, 1969.

[51] C. P. Lindsey and G. D. Patterson. “Detailed comparison of the Williams-Watts and Cole-Davidson functions”. In: *J. Chem. Phys.* 73 (1980), p. 7.

[52] A.C. Ling and J.E. Willard. “Viscosities of some organic glasses used as trapping matrixes”. In: *J. Phys. Chem.* 72.6 (1968), p. 1918.

[53] Douglas Main. *Humankind's most important material*. 2018. URL: https://www.theatlantic.com/technology/archive/2018/04/humankinds-most-important-material/557315/.

[54] Century Manufacturing. *The Lucite Story*. 2019. URL: https://centurymfg.com/the-lucite-story/ (visited on 02/19/2019).

[55] S. Martelli, N. Giovambattista Torquato, and R. Car. "Large-Scale Structure and Hyperuniformity of Amorphous Ices". In: *Phys. Rev. Let.* 119 (2017), p. 136002.

[56] A. J. Matheson. "Role of free volume in the pressure dependence of the viscosity of liquids". In: *Am. Inst. Phys.* 44 (1966), p. 695.

[57] Shiro Matsuoka. "Free Volume, Entropy, and Relaxation Phenomena". In: *Journal of Rheology* 30.4 (1986), pp. 869–876.

[58] G. E. McDuffie and M. V. Kelly. "Effect of Pressure on the Viscosity and Dielectric Relaxation Time in Glycerol". In: *J. Chem. Phys.* 41.9 (1964), p. 2666.

[59] Raymond D. Mountain. "Spectral Distribution of Scattered Light in a Simple Fluid". In: *Journal of Research of the National Bureau of Standards* 3 (1966), p. 207.

[60] Raymond D. Mountain. "Spectral Distribution of Scattered Light in a Simple Fluid". In: *Rev. Mod. Phys.* 38.1 (1966), p. 205.

[61] J. Mydosh. *Spin Glasses*. CRC Press, 1993.

[62] K. L. Nagai, P. Lunkenheimer, and C. Leon. In: *J. Chem. Phys.* 116 (2002), p. 9839.

[63] Design News. *Use of Metallic Glass Simplifies 3D Printing of Metals*. 2018. URL: https://www.designnews.com/materials-assembly/use-metallic-glass-simplifies-3d-printing-metals/2046166359487 (visited on 02/19/2019).

[64] Michael I. Ojovan and William E. Lee. "Glassy Wasteforms for Nuclear Waste Immobilization". In: *Metallurgical and Materials Transactions A* 42.4 (2011), p. 837.

[65] M. Paluch et al. "Effect of pressure on the α relaxation in glycerol and xylitol". In: *J. Chem. Phys.* 116.22 (2002), p. 9839.

[66] G. D. Patterson and D. J. Ramsay. "Depolarized light-scattering spectroscopy and polymer characterization". In: *Anal. Chim. Acta.* 189 (1986), p. 57.

[67] M. Pawlus, J. Paluch, and J. Ziolo. "On the pressure dependence of the fragility of glycerol". In: *J. Phys.: Condens. Matter* 21 (2009), p. 332101.

[68] John Plodinec. "Borosilicate Glasses for Nuclear Waste Immobilization". In: *Glass Technology - European Journal of Glass Science and Technology Part A* 41 (Dec. 2000), pp. 186–192.

[69] A. A. Pronin et al. "Pressure-Inuduced change in the relaxation dynamics of glycerol". In: *JETP Letters* 92.7 (2010), p. 479.

[70] Tim Ransom. "Viscous Liquids and the Glass Transition at Extremely High Pressure: Optical Techniques Applied to Cumene in a Diamond Anvil Cell". PhD thesis. University of Arkansas, 2016.

[71] Andreas Reiser, Gernot Kasper, and Siegfried Hunklinger. "Pressure-induced isothermal glass transition of small organic molecules". In: *Phys. Rev. B* 72 (2005), p. 94204.

[72] S. A. Rice. In: *Phys. Rev.* 112 (1958), p. 804.

[73] C. M. Roland. *Dielectric Relaxation Spectroscopy*. 2014. URL: http://polymerphysics.net/pdf/ImpedanceSpectroscopy%20chapter.pdf (visited on 07/17/2017).

[74] C. M. Roland, S. Blair, and R. Casalini. "Thermodynamic scaling of the viscosity of van der Waals, H-bonded, and ionic liquids". In: *J. Chem. Phys.* 125 (2006), p. 124508.

[75] C. M. Roland et al. "Supercooled dynamics of glass-forming liquids and polymers under hydrostatic pressure". In: *Rep. Prog. Phys.* 68.6 (2005), p. 1405.

[76] David Sherrington and Scott Kirkpatrick. "Solvable Model of a Spin-Glass". In: *Phys. Rev. Lett.* 35 (26 1975), p. 1792.

[77] W. Shockley. *Imperfections in nearly perfect crystals (see chapter by C. Zener)*. John Wiley & Sons, Inc., New York, 1952.

[78] David Sinclair. "Light scattering by spherical particles". In: *J. Opt. Soc. Am.* 37 (1947), p. 475.

[79] R. N. Sinclair. "Inelastic neutron scattering from amorphous solids". In: *J. Non-Crystalline Solids* 76 (1 1985), p. 61.

[80] Anders Sodergard and Mikael Stolt. "Properties of lactic acid based polymers and their correlation with composition". In: *Progress in Polymer Science* 27.6 (2002), p. 1123.

[81] C. Soundrapandian et al. "Porous bioactive glass scaffolds for local drug delivery in osteomyelitis: development and in vitro characterization." In: *AAPS PharmSciTech* 11 (2010), p. 1675.

[82] THE Jennifer Stabach. “Retrofitting a Monochromator for High-Pressure Raman and Optical Studies of Glass-Forming Liquids”. B.S. Thesis. University of Arkansas, 2013.

[83] J. L. Sutterby. In: *Trans. Soc. Rheol.* 17 (1973), pp. 559, 575.

[84] G. Tammann and W. Hesse. In: *Z. anorg. allgem. Chem.* 156 (1926), p. 245.

[85] Albert Tolle. “Neutron scattering studies of the model glass former ortho-terphenyl”. In: *Rep. Prog. Phys.* 64 (2001), p. 1473.

[86] P.L. Travers. *Mary Poppins*. New York: Harcourt, Brace & Co., 1934.

[87] G. H. Vineyard. “Liquid Metals and Solidification”. In: *Am. Soc. Metals, Cleveland, Ohio* (1958), p. 1.

[88] Li-Min Wang, C. Austen Angell, and Ranko Richert. “Fragility and thermodynamics in nonpolymeric glass-forming liquids”. In: *J. Chem. Phys.* 125.7 (2006), p. 74505.

[89] Ronald P. White and Jane E. G. Lipson. “Polymer free volume and its connection to the glass transition”. In: *Macromolecules* 49.11 (2016), p. 3987.

[90] G. Williams and David C. Watts. “Non-symmetrical dielectric relaxation behavior arising from a simple empirical decay function”. In: *Trans. Farad. Soc.* 66 (1970), p. 80.

[91] Malcom L. Williams, Robert F. Landel, and John D. Ferry. “The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids”. In: *J. Am. Chem. Soc.* 77 (1955), p. 3701.

[92] Peter G. Wolynes and Vassiliy Lubchenko. *Structural glasses and supercooled liquids*. John Wiley & Sons, Inc., Hoboken New Jersey, 1991.

[93] S. Yasutomi, S. Bair, and W. O. Winer. “An application of a free volume model to lubricant rheology I–Dependence of viscosity on temperature and pressure”. In: *J. Tribology* 106 (1984), p. 291.

[94] Edgar Dutra Zanotto. “Do cathedral glasses flow”. In: *Am. J. Phys.* 66 (1998), p. 5.

[95] J. Zarzycki. *Glasses and the vitreous state*. Cambridge University Press, Cambridge New York, 1991.